

# FELIX KLEIN

felixklein84.github.io

## EDUCATION

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### Technical University of Munich (TUM)

Munich

*B. Sc. Physik*

10/2025 - expected 10/2026

- Additional physics degree pursued to extend a mathematics-based analytical profile towards technical, computational and biophysical research
- Relevant focus: simulation, dynamics, applied physics and preparation for graduate study in Applied and Engineering Physics, ESPACE, Robotics, Cognition, Intelligence, or Mathematics in Data Science

### University of Mannheim

Mannheim

*B.Sc. Business Mathematics / Final grade: 2.3*

09/2020 - 09/2024

- Bachelor thesis: "Mathematical Brain Tumour Modelling Using the Example of Glioblastoma: Analysis of Cell Dynamics and Therapy Effects with Continuous-Time Markov Chains"
- Core areas: Markov chains, stochastic processes, dynamical systems, numerical methods, optimisation and scientific programming

### Western Sydney University

Sydney, Australia

*Exchange Semester / Mathematics and Computer Science / Grade: 1.7*

07/2023 - 02/2024

- Coursework in Machine Learning, Data Science and Mathematical Modeling
- Project work on convolutional neural networks for image classification and feature-map visualisation using PyTorch

### Rheingau Gymnasium

Geisenheim

*German Abitur / Final grade: 1.9*

07/2012 - 06/2020

- Awarded the DPG Physics Abitur Prize

## PRACTICAL AND PROFESSIONAL EXPERIENCE

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### Deutsche Aircraft

Oberpfaffenhofen

*Working Student, Supply Chain and Digitalisation*

02/2026 - present

- Modelling and optimisation of data flows between ERP and the SupplyOn supplier platform for the serial production ramp-up of the D328eco
- Identification of systematic bottlenecks in supplier communication and derivation of quantitative metrics for data quality, order confirmations and delivery-date reliability

### Technical University of Munich - Chair of Agricultural Mechatronics

Garching

*Student Research Assistant (HiWi), Collaborative Routing and Simulation*

02/2026 - present

- Literature review on collaborative multi-agent routing algorithms and Controlled Traffic Farming as a combinatorial optimisation problem
- Development of a Python simulation environment for evaluating route-allocation and consensus algorithms on realistic agricultural field graphs
- Implementation and performance evaluation of collaborative routing heuristics with respect to runtime, solution quality and scalability; reproducible documentation with Sphinx

### BMW Group – Research and Innovation Centre

Munich

*Intern, Logistics and Supply Chain Management*

04/2025 - 09/2025

- Developed a forecasting pipeline for axle-assembly demand in prototype production using Python, time-series modelling and Power BI visualization
- Designed a discrete-event simulation model for warehouse utilisation at the Parsdorf supply centre, quantifying bottlenecks in inventory and disposal logistics

- Supported the NAO VS0 8/26 prototype cubing process through CATIA V5 and VisView geometry preparation, tolerance analysis and coordination between engineering, quality and suppliers
- Documented quality deviations in PQM Next in German and English, creating a structured data basis for statistical analysis in prototype testing

### **KombiRail Europe B.V.**

*Intern, Executive & Operations*

Frankfurt

01 - 31/03/2025

- Prepared quantitative analyses of operational and financial performance indicators for senior management
- Modelled cost-saving scenarios of approximately EUR 400k p.a. as decision support for an M&A-related process

### **KPMG - Lighthouse**

*Intern, Data Science Consulting*

Saarbrücken

09/2024 - 02/2025

- Extended the Python backend of a simulation software with numerical analysis modules; vectorisation and caching reduced runtime of complex scenarios by approximately 25%
- Automated UI regression tests for an Angular frontend with Selenium and integrated them into a Git/Azure DevOps CI pipeline, reducing manual testing effort by around 50% per sprint
- Refactored and optimised R scripts for stochastic reserve forecasting, supporting quantitative risk assessments for client projects

### **Cosuno Ventures GmbH**

*Working Student, Sales Operations*

Berlin

08/2021 - 09/2022

- Designed automated reporting workflows using HubSpot, Excel/VBA and data-cleaning routines, reducing KPI report preparation from two days to three hours
- Supported weekly data-driven management updates and operational sales controlling in a venture-backed construction-tech start-up

### **Family Winery Martin Klein**

*Operations, E-Commerce and Marketing Support*

Johannisberg

01/2015 - present

- Contributed across operations, sales and customer service in the family winery, providing long-term practical exposure to logistics, hospitality and small-business management
- Built and maintained an e-commerce and digital-marketing channel using WooCommerce, SEO, web analytics and conversion-data evaluation

## **ACADEMIC AND APPLIED PROJECTS**

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### **Hackathons & Datathons**

*Participant and Team Contributor*

Hamburg & Mannheim

2023 & 2024

- 2nd place, ChefTreff Hackathon 2024 with Lufthansa Industry Solutions and ZeroG: stochastic forecasting pipeline using XGBoost and Monte Carlo simulation to optimise onboard catering under demand uncertainty
- STADS Datathon 2023, EY Challenge: Python-based matching algorithm for detecting duplicate financial transactions using fuzzy hashing and feature engineering; fraud heatmap visualisation for case-based savings analysis

## **ENGAGEMENT AND LEADERSHIP**

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### **HORYZN Student Aerospace Initiative**

*Aerostructures Team - Landing Gear*

Munich

01/05 - present

- Responsible for the development and optimisation of landing-gear concepts for autonomous aircraft prototypes
- Analysing landing stability, shock absorption, structural constraints and manufacturability; supporting CAD prototyping and interdisciplinary aerospace design work

**Volunteer Teacher – Indian High School, Jaipur***Afternoon Programme, approximately five days per week*

Jaipur, India

18/08 - 31/09/2025

- Designed and taught a practical “English for Job Interviews” course for students in grades 10-12, including structured speaking practice and simulated interviews
- Developed a “Digital Literacy and AI” module in which students created CVs, used generative AI for guided research and built small presentation-based or programming-oriented mini projects

**TG Rüdesheim „JuSki“***Snowboard Instructor*

Mühlbach am Hochkönig, Austria

01/2024 &amp; 01/2025

- Instructor for approximately 50 teenagers aged 14-17 as part of a 12-person supervision and coaching team

**STADS – Student Association for Data Analytics & Statistics***Member*

Mannheim

09/2022 - 09/2025

- Participated in more than 20 workshops and guest lectures on data science, statistics and machine learning with companies including SAP and Mercedes-Benz

**Student Council Mathematics - University of Mannheim***Quality Assurance and Budget Allocation*

Mannheim

09/2022 - 09/2023

- Contributed to the allocation of an annual student-body budget of approximately EUR 500k and supported a transparent, department-based prioritisation process

**SKILLS**

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Languages: German (native) · English (C2) · Spanish (B1) · Japanese (A2)

Programming and data science: Python (NumPy, Pandas, Scikit-Learn, PyTorch) · Julia · R · MATLAB · basic C++

Mathematical methods: Markov chains · stochastic processes · dynamical systems · numerical optimisation ·

Monte Carlo simulation · machine learning

Tools: Git · LaTeX · Sphinx · Power BI · Azure DevOps CI/CD · CATIA V5 · VisView · HubSpot · Salesforce

Interests: Applied physics · automation · endurance running · snowboarding · surfing and hands-on technical prototyping